

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

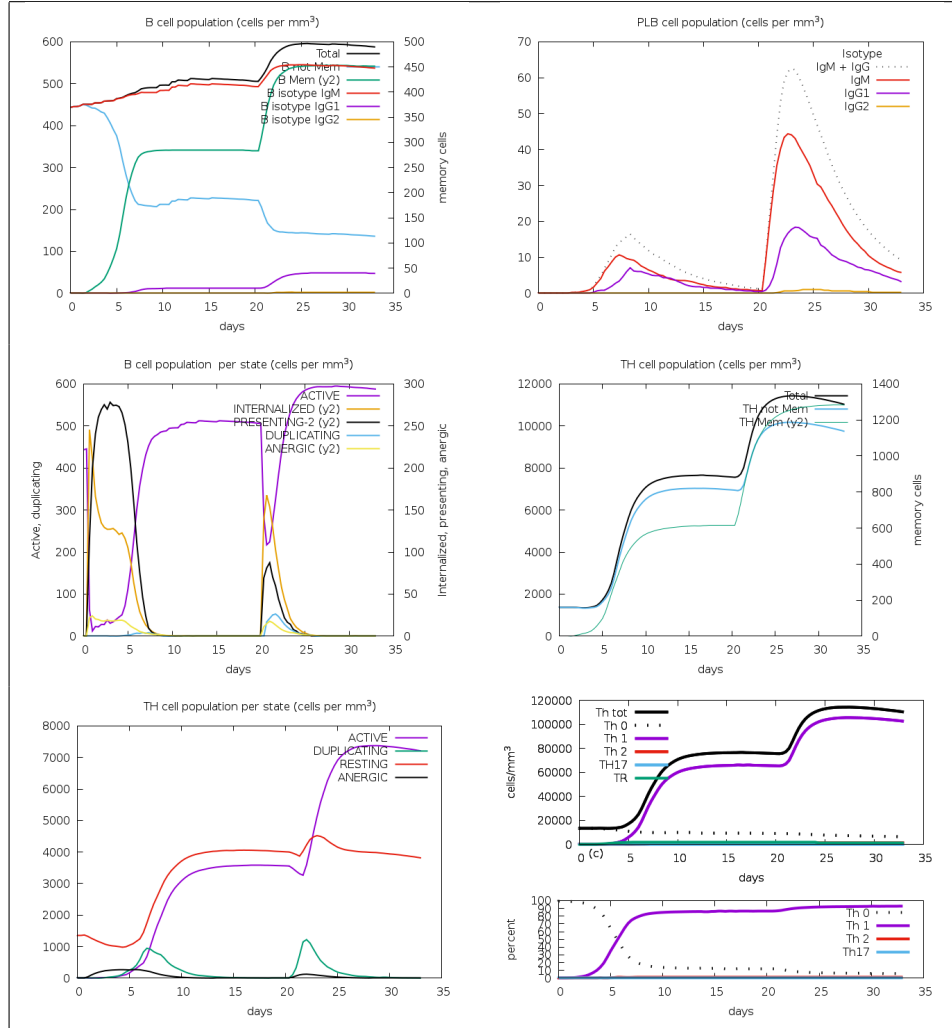


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

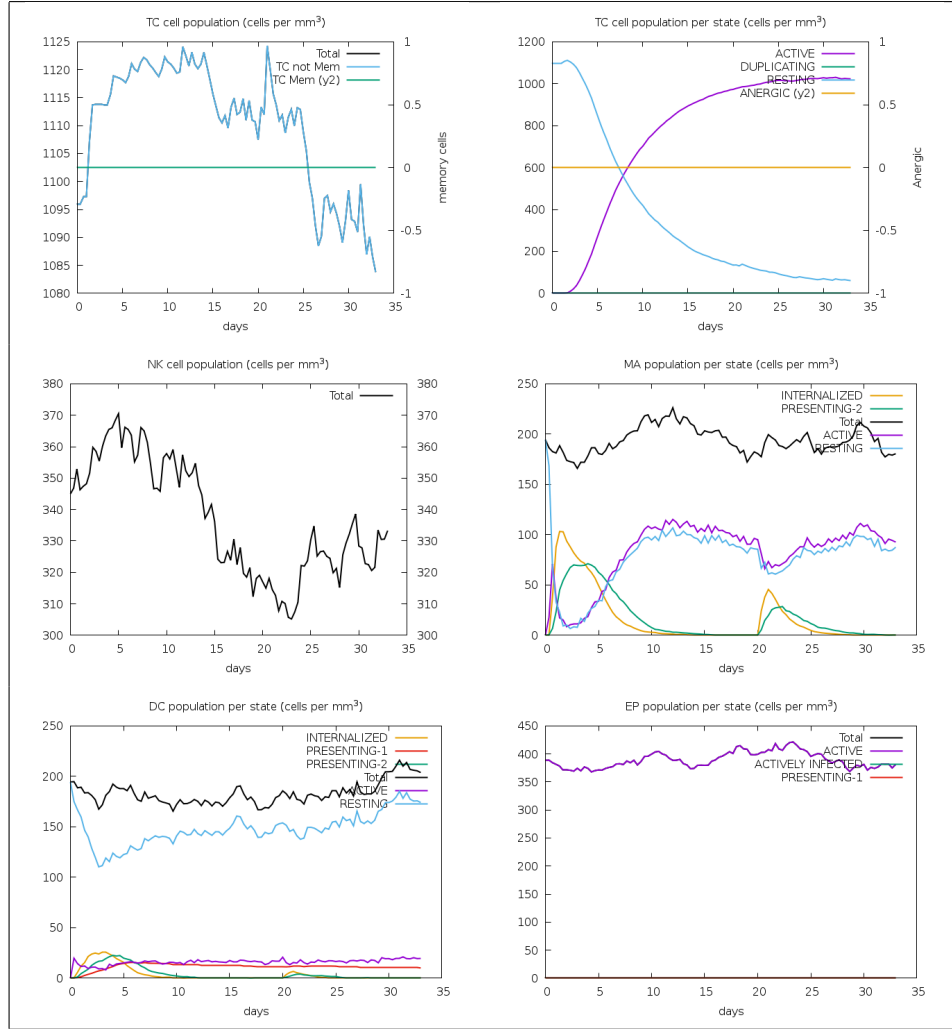


Figure 2: Legend: symbols as figure above.

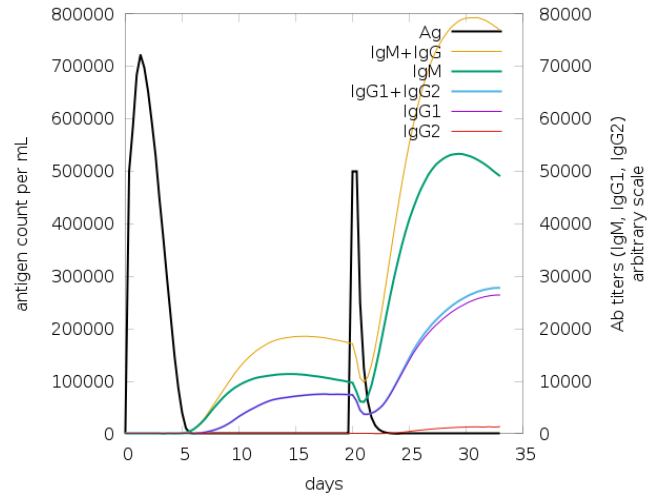


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

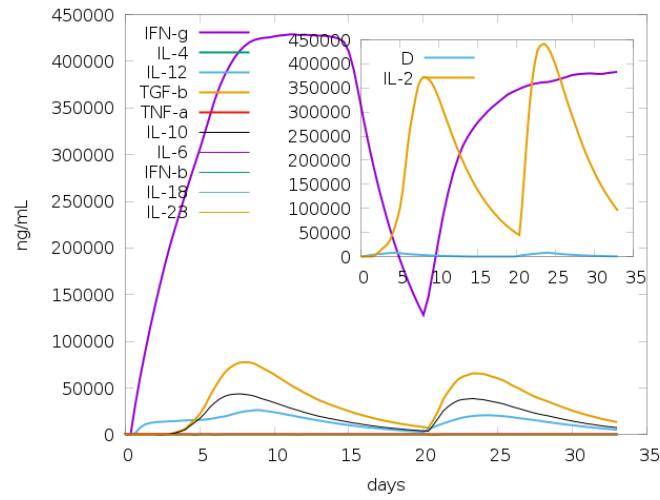


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Parker_B_scale_epitope_find
Use Parker's propensity scale, takes an antigen block as input,
and creates a list of residues that are possible epitopes.

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015144_5_OzOgADJQUHXh.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEGDTSGPEGSGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGAGAGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGT
TLGPGPGPLEKQLFYIYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALYEASTGPG

00000000010000000000000100111111000111111111111111111111111111111111111111
11111101111001110000000111
111111111100
00
00

Epitopes of protein 0 -----
1] pos=25 len=6 GPGTGP
2] pos=34 len=53 LGEKGDTSGPEGSGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGGSGSGPRH
3] pos=88 len=4 DGVR
4] pos=104 len=66 GTHGGTGAGAGAGGAGAGAGAGGGAGAGGGAGGAGGAGAGAGGGAGAGGGAGGAGAGGGAG
5] pos=279 len=19 AAEGDDGDGPGPGSDREVR
6] pos=305 len=5 STGPG

DoPeptideList_I:
Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 4 MHC I molecules

Read class I peptide list from file? NO

=====

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015144_5_OzOgADJQUHXh.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEGDTSGPEGSGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGGAGGAGGAGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGT
TLGPGPGPLEKQLFYIYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALYEASTGPG

Epitopes of protein 0 -----
0] pos= 247 score=0.033756 unnormalised=3.6626000000 PLEKQLFYIY
1] pos= -1 score=0.966244 unnormalised=104.8410000000 non-binding event

=====

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA

Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015144_5_0z0gADJQUHXh.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGKEKDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGAGAGGAGGAGGAGGAGAGGAGAGGAGAGGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGPLEKQLFYIYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALYEASTGPG

Epitopes of protein 0 -----

0]	pos= 247	score=0.033756	unnormalised=3.6626000000	PLEKQLFYIY
1]	pos= -1	score=0.966244	unnormalised=104.8410000000	non-binding event

=====

Allele: B0702
Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015144_5_0z0gADJQUHXh.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGKEKDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGAGAGGAGGAGGAGGAGAGGAGAGGAGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGPLEKQLFYIYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALYEASTGPG

Epitopes of protein 0 -----

0]	pos= 13	score=0.005907	unnormalised=0.6622000000	LPQQQLTEA
1]	pos= 73	score=0.002687	unnormalised=0.3012000000	RPGAPGGSG
2]	pos= 92	score=0.017237	unnormalised=1.9322000000	RPQKRPSCI
3]	pos= 216	score=0.013980	unnormalised=1.5672000000	GPGLPQQQL
4]	pos= 260	score=0.024935	unnormalised=2.7952000000	LPNTRPHEA
5]	pos= -1	score=0.935254	unnormalised=104.8410000000	non-binding event

=====

Allele: B0702
Pseudo sequence: KAAREEQIKAQTRE
Threshold: 8.702800
Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015144_5_0z0gADJQUHXh.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGKEKDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGAGAGGAGGAGGAGGAGGAGAGGAGAGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGPLEKQLFYIYIGTMLPNTRPHEAAAKPPMVEGAAAEAGDDGDGPGPGSDREVRGSALYEASTGPG

Epitopes of protein 0 -----

0]	pos= 13	score=0.005907	unnormalised=0.6622000000	LPQQQLTEA
1]	pos= 73	score=0.002687	unnormalised=0.3012000000	RPGAPGGSG
2]	pos= 92	score=0.017237	unnormalised=1.9322000000	RPQKRPSCI
3]	pos= 216	score=0.013980	unnormalised=1.5672000000	GPGLPQQQL
4]	pos= 260	score=0.024935	unnormalised=2.7952000000	LPNTRPHEA

5] pos= -1 score=0.935254 unnormalised=104.8410000000 non-binding event

DoPeptidelist_II:

Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 2 MHCII molecules

Read class II peptide list from file? NO

Allele: DRB1_0101
Pseudo sequence: KFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015144_5_OzOgADJQUHXh.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGGAGAGGAGGAGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGPLEKQLFYIYIGTMLPNTRPHEAAAKPPMVEGAAEGDDGDGPGPGSDREVRGSALYEASTGPG

Epitopes of protein 0 -----

0]	pos=	5	score=0.039718	unnormalised=3.1995600000	LSGGSPEPL
1]	pos=	181	score=0.020775	unnormalised=1.6735600000	VLLGPGPG
2]	pos=	182	score=0.010571	unnormalised=0.8515600000	VLLGPGPGT
3]	pos=	198	score=0.036366	unnormalised=2.9295600000	FGPGPGACM
4]	pos=	211	score=0.011067	unnormalised=0.8915600000	LEKGPGL
5]	pos=	253	score=0.035174	unnormalised=2.8335600000	FYIYIGTMLP
6]	pos=	256	score=0.024958	unnormalised=2.0105600000	IGTMLPNTR
7]	pos=	-1	score=0.821372	unnormalised=66.1680000000	non-binding event

Allele: DRB1_0101
Pseudo sequence: KFAHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015144_5_OzOgADJQUHXh.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGGAGAGGAGGAGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGPLEKQLFYIYIGTMLPNTRPHEAAAKPPMVEGAAEGDDGDGPGPGSDREVRGSALYEASTGPG

Epitopes of protein 0 -----

0]	pos=	5	score=0.039718	unnormalised=3.1995600000	LSGGSPEPL
1]	pos=	181	score=0.020775	unnormalised=1.6735600000	VLLGPGPG
2]	pos=	182	score=0.010571	unnormalised=0.8515600000	VLLGPGPGT
3]	pos=	198	score=0.036366	unnormalised=2.9295600000	FGPGPGACM
4]	pos=	211	score=0.011067	unnormalised=0.8915600000	LEKGPGL
5]	pos=	253	score=0.035174	unnormalised=2.8335600000	FYIYIGTMLP
6]	pos=	256	score=0.024958	unnormalised=2.0105600000	IGTMLPNTR

7] pos= -1 score=0.821372 unnormalised=66.1680000000 non-binding event
